



SMART WI-FI NETWORK ANALYZER

THE PROJECT IS IN UNDER SUBJECT OF :-

DESIGN ENGINEERING – II B (3160001)

BE IT SEM: 6

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CERTIFICATE

This is to certify that the research work embodied in this report, entitled “**Smart Wi-Fi Network Analyzer,**” has been carried out by **Panchal Nirav Rajubhai** (Enrolment No. 231250116034) at **Shree Swaminarayan Institute of Technology (125)** for the partial fulfilment of the subject **Design Engineering – II B** in the **Bachelor of Engineering (Information Technology)** program to be awarded by **Gujarat Technological University**.

This research work has been carried out under my supervision and is found satisfactory as per the requirements of the department.



Internal Guide

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LIST OF FIGURES

Figures No.	Figure Description	Page no.
3.1	AEIOU CANVAS	3
3.2	MIND MAP CANVAS	5
4.1	EMPATHY CANVAS	6
5.1	IDEATION CANVAS	8
6.1	LNМ CANVAS	11
7.1	PD CANVAS	13
8.1	DASHBOARD	16
8.2	NETWORKS	16
8.3	SPEED TEST	17
8.4	SIGNAL	17
8.5	SETTINGS	18

INDEX

		Page no.
Chapter: 1	Introduction	1
Chapter: 2	Problem Definition/Project Title	2
	2.1 Domain Selection	
	2.2 User Study	
Chapter: 3	AEIOU & Mind Map Canvas	3
	3.1 AEIOU Canvas	
	3.2 Mind Map Canvas	
Chapter: 4	Empathy Canvas	6
Chapter: 5	Ideation Canvas	8
Chapter: 6	Learning Needs Matrix Canvas	11
Chapter: 7	Product Development Canvas	13
Chapter: 8	Prototype	16
Chapter: 9	Testing & Validation	19
Chapter: 10	Final Solution	20
Chapter: 11	Conclusion	26

CHAPTER:1

INTRODUCTION

With the rapid growth of wireless communication, Wi-Fi has become an essential part of daily life for activities such as browsing, streaming, online learning, and cloud-based services. However, users often face issues like slow internet speed, unstable connections, and lack of detailed information about their network. Most devices provide only basic indicators such as signal bars, which are insufficient for understanding actual network performance or diagnosing problems effectively.

To address these challenges, the Smart Wi-Fi Network Analyzer is developed as a comprehensive mobile application that provides real-time insights into Wi-Fi network performance. The application allows users to monitor signal strength, view detailed network information, analyse nearby networks, and measure internet speed through an integrated interface. By combining these functionalities into a single platform, the system eliminates the need for multiple tools and simplifies network analysis for users.

The project focuses on delivering accurate data, intuitive visualization, and a user-friendly experience. Features such as live signal graphs, speed test analysis, and historical data tracking help users better understand their network behaviour and make informed decisions to improve connectivity. Overall, this application aims to bridge the gap between basic network indicators and advanced diagnostic tools by providing a practical and efficient solution for Wi-Fi monitoring and analysis.

CHAPTER: 2

PROBLEM DEFINATION / PROBLEM TILTE

2.1 DOMAIN SELECTION

The domain selected for this project is **Computer Networks and Mobile Application Development**, with a specific focus on **network monitoring and analysis tools**. This domain was chosen due to the increasing dependence on wireless networks and the growing need for efficient tools to monitor and troubleshoot Wi-Fi performance in real time.

With the expansion of digital services such as online education, cloud computing, video streaming, and remote work, reliable network connectivity has become critical. However, most users lack access to simple yet powerful tools that can provide detailed insights into their network behavior. Existing solutions are often either too basic or overly complex, creating a gap between usability and functionality.

By working in this domain, the project leverages core concepts of computer networks such as signal strength (RSSI), IP addressing, latency (ping), bandwidth measurement, and wireless communication standards. At the same time, it integrates these concepts into a mobile application environment, ensuring accessibility and ease of use for end users.

Thus, the selected domain not only aligns with current technological trends but also addresses real-world problems, making the Smart Wi-Fi Network Analyzer a practical and relevant solution in the field of network analysis and mobile computing.

2.2 USER STUDY

A brief user study was conducted by observing common issues faced by Wi-Fi users, including slow internet speed, unstable connections, and lack of clear network information. Most users rely on basic indicators such as signal bars, which do not provide detailed insights into actual network performance. Additionally, existing tools are often complex or limited to single functionalities, making them less user-friendly for everyday use. This study highlights the need for an integrated and simple tool that can provide real-time information such as signal strength, IP details, ISP data, and speed performance in one place. Therefore, the Smart Wi-Fi Network Analyzer is developed to address these gaps by offering a user-friendly interface and comprehensive network analysis, helping users better understand and troubleshoot their Wi-Fi connectivity.

CHAPTER: 3

AEIOU & MIND MAPPING CANVAS

3.1 AEIOU CANVAS

- AEIOU canvas includes Activities, Users, Objects, Interactions, and Environment.

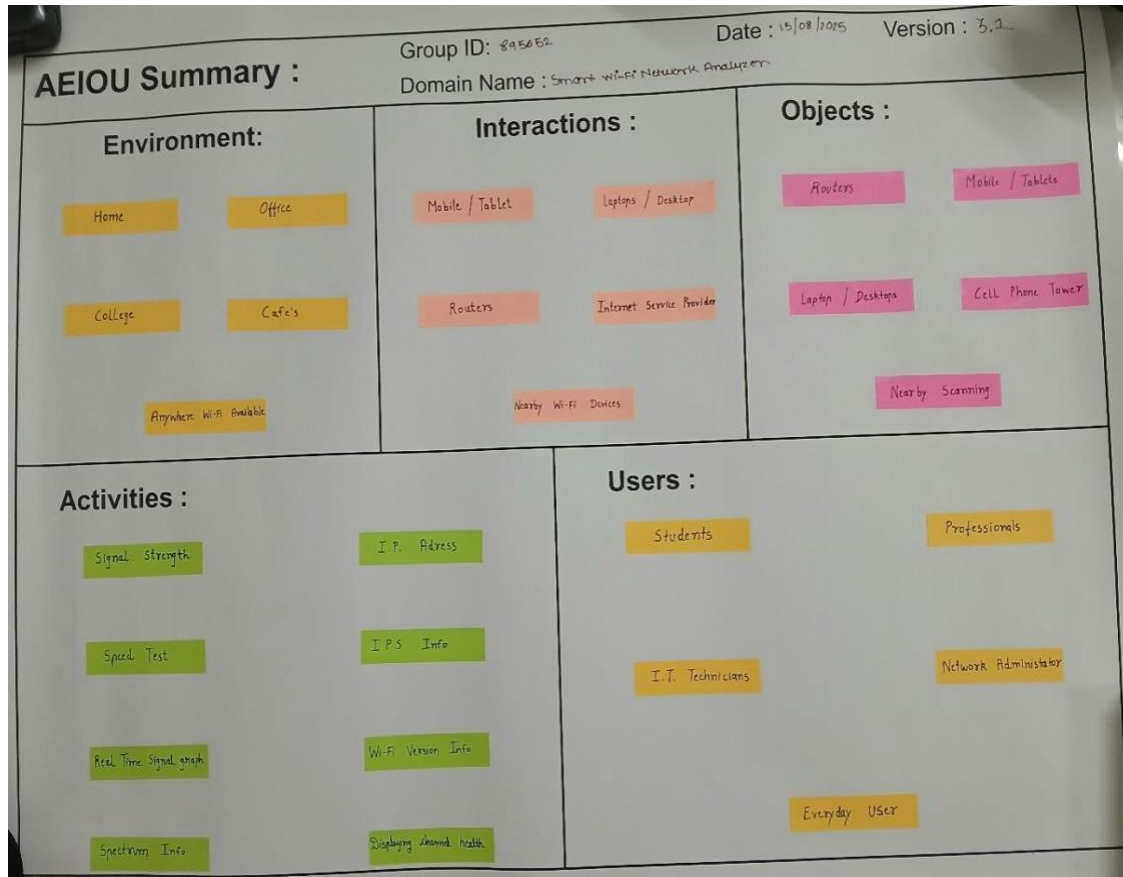


Figure 3.1. AEIOU Canvas

Environment:

- Home.
- Office.
- College.
- Cafés
- Anywhere Wi-Fi available.

- Mobile/Tablet.
- Laptop/Desktop.
- Routers.
- Internet service provider.
- Nearby available Wi-Fi devices.

Objects:

- Routers.
- Mobile/Tablets.
- Laptop/Desktop.
- Cell phone towers.
- Nearby scanning.

Activities:

- Signal strength.
- Internet speed test.
- Real time signal graph.
- Spectrum info.
- IPS info.
- IP address.
- Displaying channel health.
- Wi-Fi version info.

Users:

- Students.
- Professionals.
- IT technicians.
- Network administrators.
- Everyday user.

3.2 MIND MAPPING CANVAS

- Mind Mapping Canvas includes the overall structure of App.

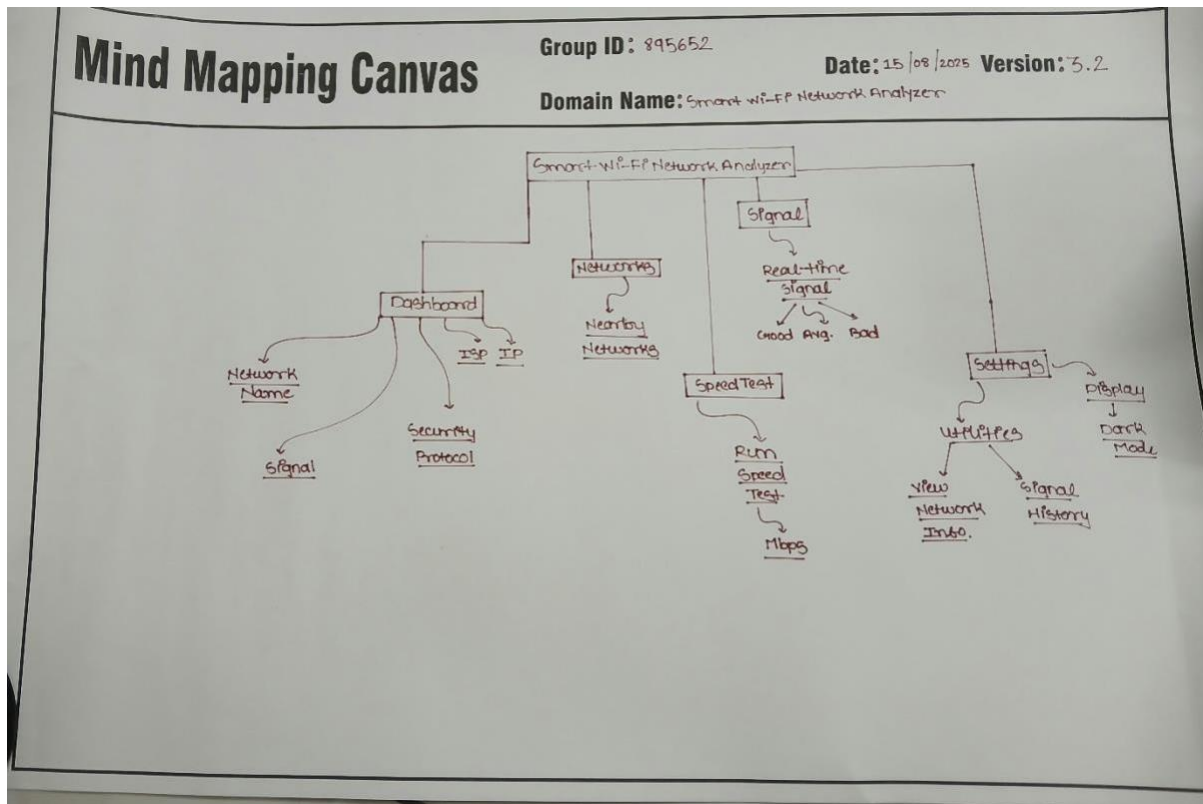


Figure 3.2. Mind Mapping Canvas

CHAPTER: 4

EMPATHY CANVAS

- Empathy canvas has information like Users, Stakeholders, Activities, and Storyboarding.

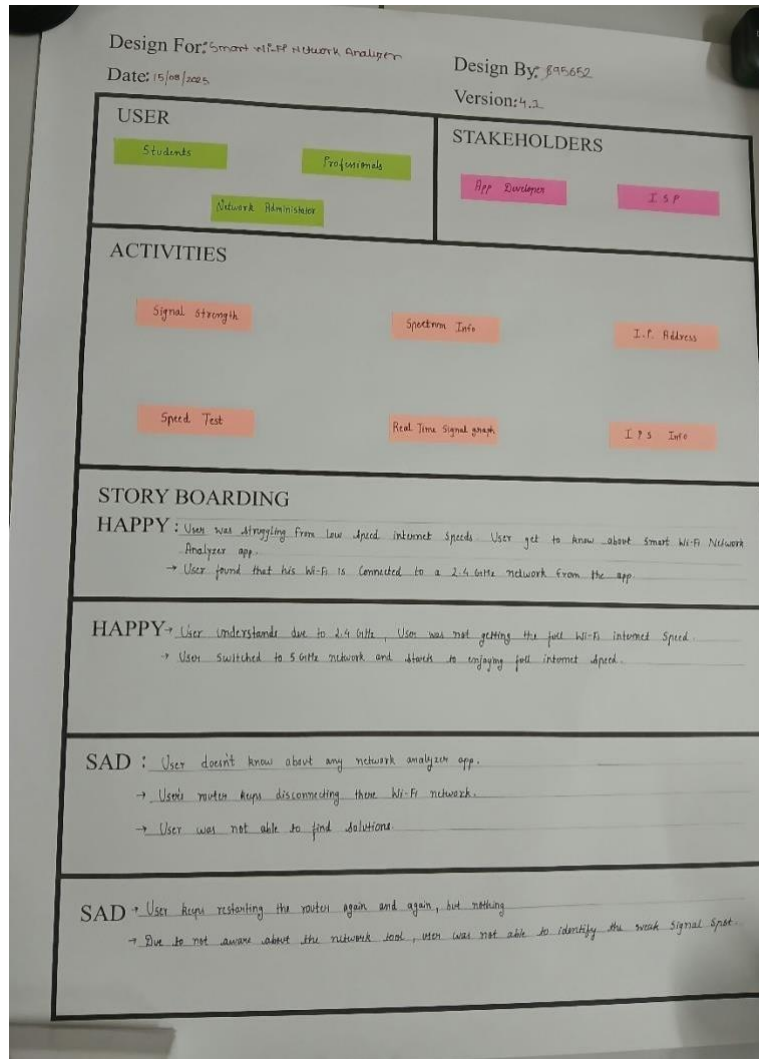


Figure 4.1. Empathy Canvas

Users:

- Students.
- Professionals.
- IT technicians.
- Network administrators.
- Everyday user.

Stakeholders:

- App developer.
- ISP.

Activities:

- Signal strength.
- Internet speed test.
- Real time signal graph.
- Spectrum info.
- ISP info.
- IP address.
- Displaying channel health.
- Wi-Fi version info.

Story boarding:

Happy

- User was struggling from low speed internet speeds.
- User get to know about smart wi-fi network analyzer app.
- User found that his Wi-Fi is connected to a 2.4 GHz network from the app.
- User understands due to 2.4 GHz, user was not getting the full wi-fi internet speeds.
- User switched to 5 GHz network and starts to enjoying full internet speeds.

Sad

- User doesn't know about any network analyzer app.
- User's router keeps disconnecting there wi-fi network.
- User was not able to find solutions.
- User keeps restarting the router again and again, but nothing.

CHAPTER: 5 IDEATION CANVAS

- Ideation Canvas includes People, Activities, Situation/Context/Location, Props/Possible Solution.

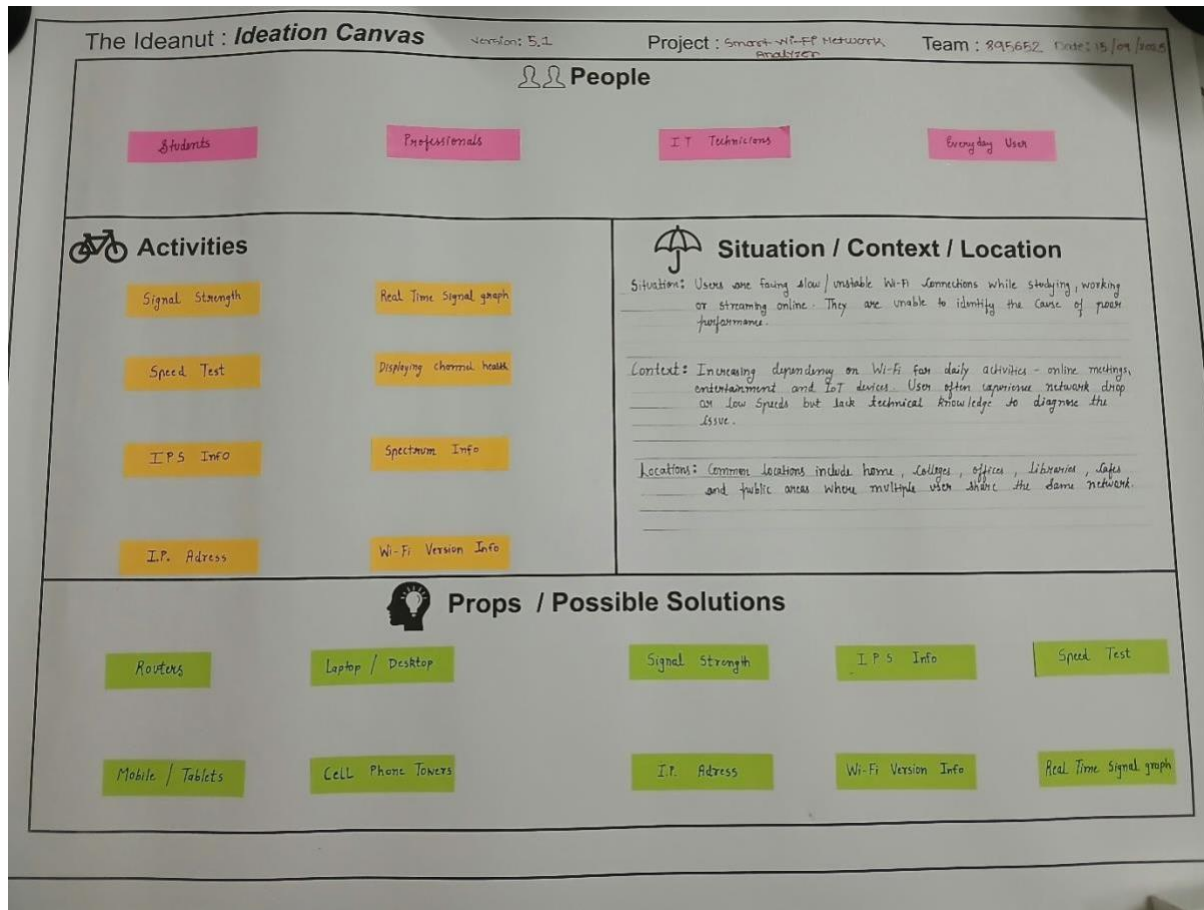


Figure 5.1. Ideation Canvas

People:

- Students.
- Professionals.
- IT technicians.
- Network administrators.
- Everyday user.

Activities:

- Signal strength.
- Internet speed test.
- Real time signal graph.
- Spectrum info.
- IPS info.
- IP address.
- Displaying channel health.
- Wi-Fi version info.

Situation/context/location:

Situation

- Users are facing slow or unstable Wi-Fi connections while studying, working, or streaming online. They are unable to identify the cause of poor performance.

Context

- Increasing dependency on Wi-Fi for daily activities — online classes, meetings, entertainment, and IoT devices. Users often experience network drops or low speeds but lack technical knowledge to diagnose the issue.

Locations

- Common locations include homes, college hostels, offices, libraries, cafes, and public Wi-Fi areas where multiple users share the same network.

Props/possible solution:

Props

- Routers.
- Mobile/Tablets.
- Laptop/Desktop.
- Cell phone towers.

- ISP.

Possible solutions:

- Signal strength.
- Internet speed test.
- Real time signal graph.
- Spectrum info.
- IPS info.
- IP address.
- Displaying channel health.
- Wi-Fi version info.

CHAPTER: 6

LEARNING NEEDS MATRIX CANVAS

- Learning Needs Matrix Canvas includes Tools/Methods/Theory Application Process Involved, Application standard and design specification/Principles and Experiments, Software Simulation/Skill/Mathematical Requirement, Component materials strength criteria

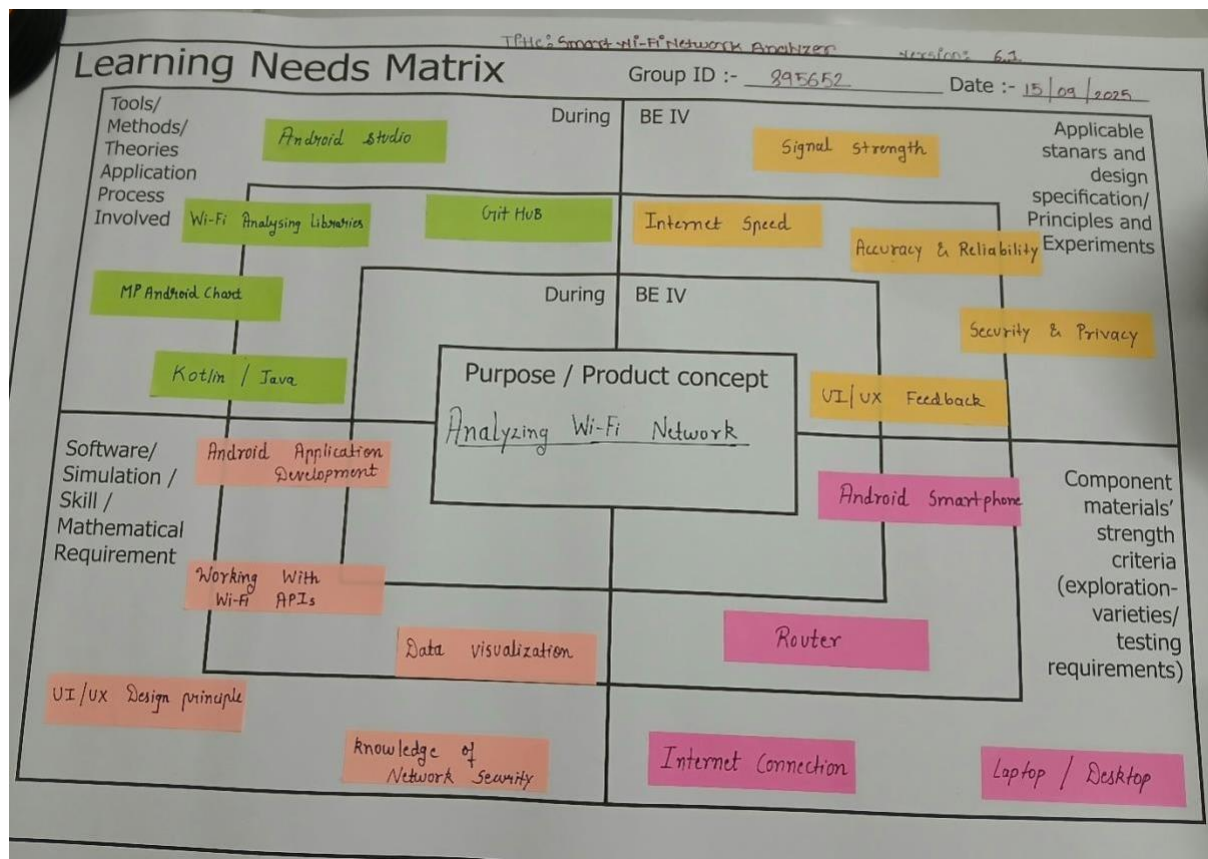


Figure 6.1. Learning Needs Matrix Canvas

Purpose:

- Analyzing Wi-Fi network.

Tools:

- Android Studio.
- Kotlin/Java.
- Wi-Fi analyzer libraries.
- MP android chart/graph view.
- Firebase/local storage.
- GitHub.

Skills:

- Android app development.
- Working with Wi-Fi APIs.
- Data visualization and chart integration.
- UI/UX design principles.
- Basic knowledge of network security.

Experiments:

- Signal strength measurement.
- Internet speed test.
- Accuracy & reliability.
- Security & privacy.
- UI/UX feedback.

Components:

- Android smartphone.
- Router.
- Internet connection.
- Laptop/Desktop.

CHAPTER: 7

PRODUCT DEVELOPMENT CANVAS

- Product Development Canvas includes Purpose, Product Experience, Product Functions, Product Features, Components, Customer Revalidation, Reject, Redesign, Retain and People.

Product Development Canvas

Team: Smart Wi-Fi Network Analyzer ID: 895652 Date: 15/09/2025 Version: 7.1

① Purpose What is the purpose of this canvas you're developing? Does it solve a problem, or it enhance a certain experience? Is it something a need or it is trying to create a new need or top an unmet need? Identify Wi-Fi Issues Analyze Network Security Educational Tool Assist Different User	⚓ Product Experience Define what your customer should feel like when he uses your product / service? Emotions, feelings would define his experience? Feeling Convenience, or feeling of buying more with less (cost conscious) or feeling of greater security, safety etc. Clear Interface Live Data Refresh Analyse & optimization	☑ Customer Revalidation Once you're finished with your feature set, test with the customer / user if the features, functions are useful. Speak to the customer / user. UI is clean & Simple Understandable by non-tech person Which channel is best for button output User can check how strong is Wi-Fi
	⊗ Product Functions Functions are a products answer to user problems / need. They do something that user wants. They are often verbs in nature. Every function is powered by many features. Multitasking is a function. Browser tabs is a feature that powers the multitasking function. A function can have one or more features powering it. Functions are very generic in nature. Features are often more specific. Functions can be similar to product experience. Safety (product function) provides a feeling of safety (product experience). Network Security Scan and detect nearby Networks Real Time Signal graph	
👤 People Who is the key customer segment who will use this product / service or the end product of the canvas you're pursuing? Write here about them, describe them a little. Students IT Technicians Professionals Network Administrator	⊕ Product Features Product features are specific. One or more features will power a function. Android Brakes, Airbags are feature that power the safety function. Browser tabs, Apple's Home Button to multitask between apps are features powering the multitasking function. Each feature will have many component-level components powering it. Sometimes a very popular component becomes a feature in itself. Like car stereo is a major components and a feature at the same time powering the in car entertainment function powering entertainment as a product experience. User-friendly Interface Speed Test Channel Interface Signal strength ISP Info	🔧 Reject, Redesign, Retain Post customer validation, reject those function or feature that the customers didn't find useful. Redesign those that were partially useful and retain those that the bar. Iterate with this until all functions / features are accepted. Real time Signal strength UI / UX Security Analysis App stability Connected device list
	⚙️ Components Components build up the features. For a airbag it will comprise a lot of component like bags, triggers etc. that go into making it. For a tabbed browser it will comprise of various chunks of code that will make the tabs work. In cases where the feature is a major component, you could list here the auxiliary components that are required to make the major component work. You can also list how adjustments and innovations you're planning here at the component level. Android Smartphone Router Internet connection Laptop / Desktop	

Figure 7.1. Product Development Canvas

Purpose:

- Identify Wi-Fi issues quickly.
- Optimize network performance.
- Analyze network security.
- Assist different users.
- Educational & practical tool.

Product experience:

- Provides a smooth and intuitive way for users to analyze, optimize, and secure wi-fi networks with real-time insights and actionable recommendations.

Product functions:

- Scan and detect nearby networks.
- Real-time signal strength.
- Network security.
- Wi-fi version.
- Wi-fi signal graph.

Product features:

- Signal strength.
- Internet speed test.
- Wi-Fi standards info.
- Channel interference.
- User-friendly dashboard.
- ISP info.
- Spectrum info.

Components:

- Android smartphone.
- Router.
- Internet connection.
- Laptop/Desktop.

Customer revalidation:

- Test the app with students, office workers, and home users.
- Observe how easily users scan networks and check wi-fi performance.

- Collect feedback on app usability, visuals, and recommendation clarity.
- Make iterative improvements based on user suggestions and difficulties.
- Ensure the app effectively solves wi-fi problems and enhances user satisfaction.

Retain:

- Realtime Wi-Fi signal strength.
- Security analysis.
- Smart suggestion.
- Connected device list.

Reject:

- Features that users rarely used or found unnecessary.

Redesign:

- UI.
- UX..
- App stability.
- Features.

CHAPTER: 8

PROTOTYPE

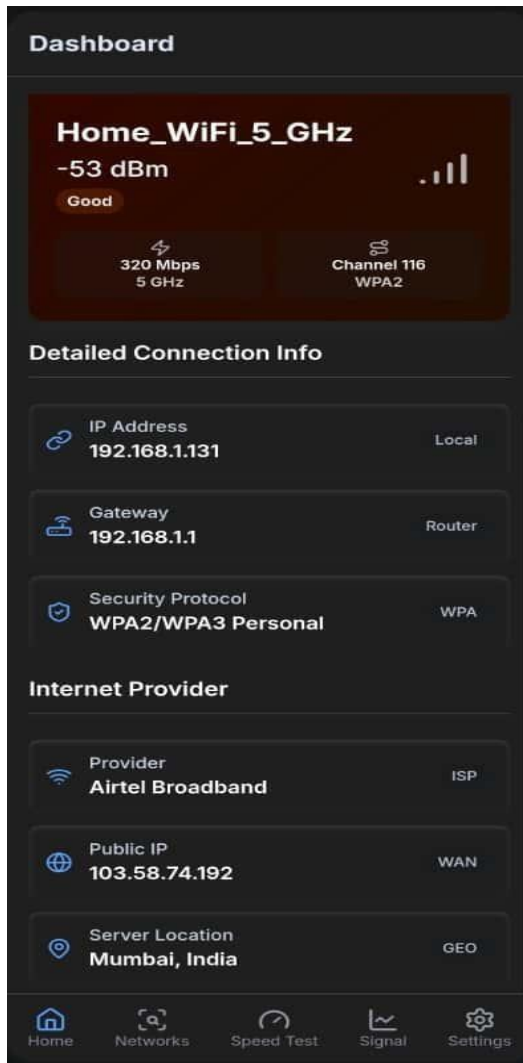


Figure 8.1. Dashboard

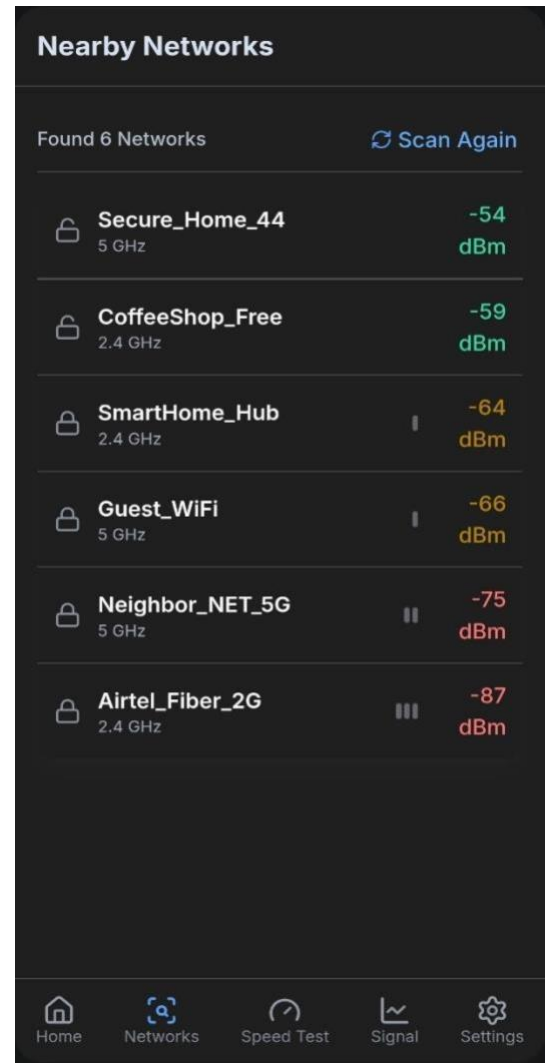


Figure 8.2. Network



Figure 8.3. Speed Test

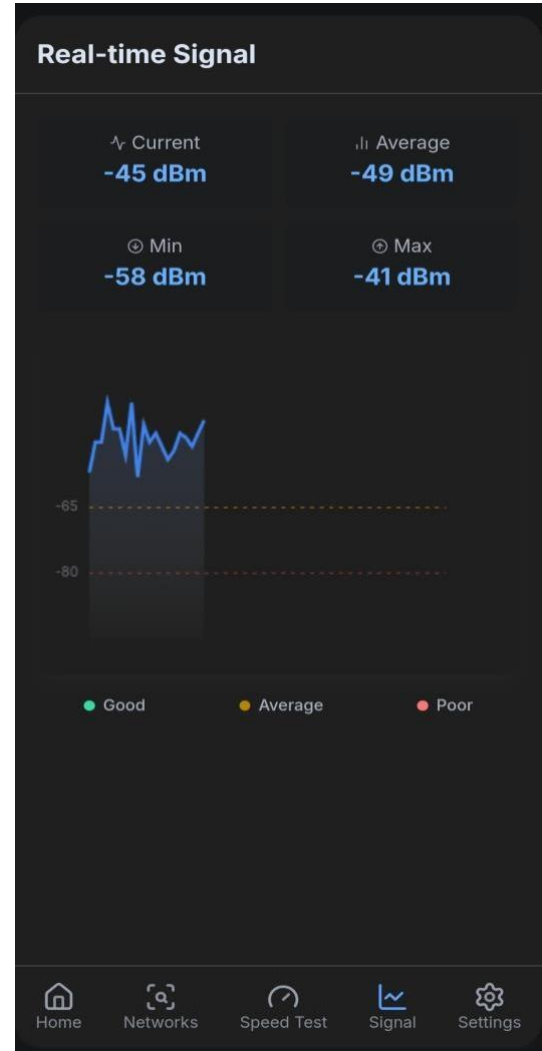


Figure 8.4. Signal

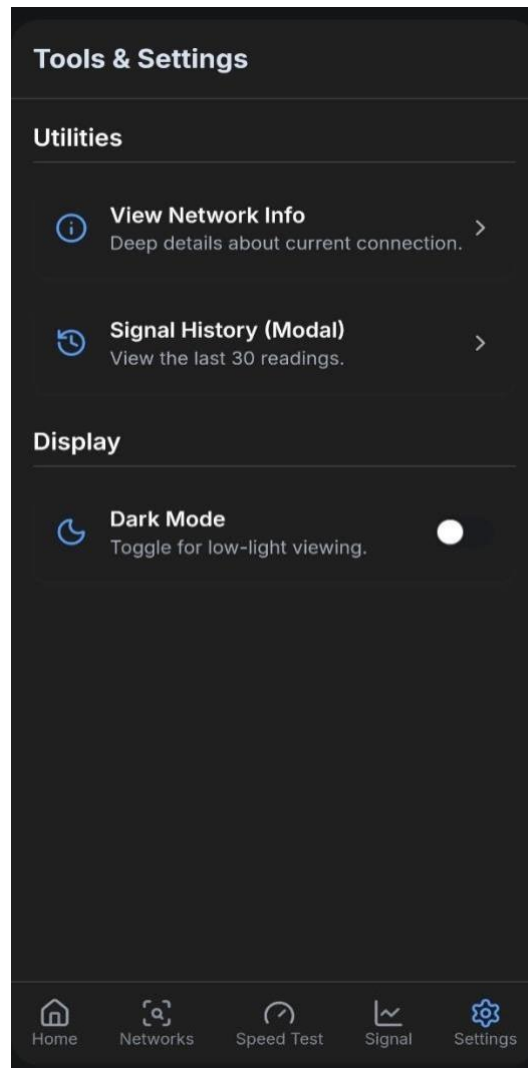


Figure 8.5. Settings

CHAPTER: 9

TESTING & VALIDATION

The testing and validation phase of the Smart Wi-Fi Network Analyzer was conducted to ensure that the application performs accurately, reliably, and efficiently under different network conditions. Various types of testing were carried out, including functional testing, performance testing, accuracy testing, and user interface testing. Functional testing confirmed that all core modules such as Wi-Fi information display, IP address detection, ISP identification, signal strength monitoring, speed testing, and history tracking were working as expected. Performance testing was conducted under different network conditions, including high-speed, moderate-speed, and weak signal environments, and the application demonstrated stable performance without crashes or major delays.

Accuracy testing was performed by comparing the application's output with standard tools such as Speed test by Ookla and browser-based speed test services. The results showed that parameters like signal strength, IP address, and ISP information were highly accurate and consistent with external references. However, the speed test module displayed slightly lower values compared to actual network speed, typically around 50–70% of the expected value. This deviation is primarily due to limitations in implementation, such as fewer parallel connections and non-optimized server selection, unlike professional tools that use advanced multi-threaded techniques.

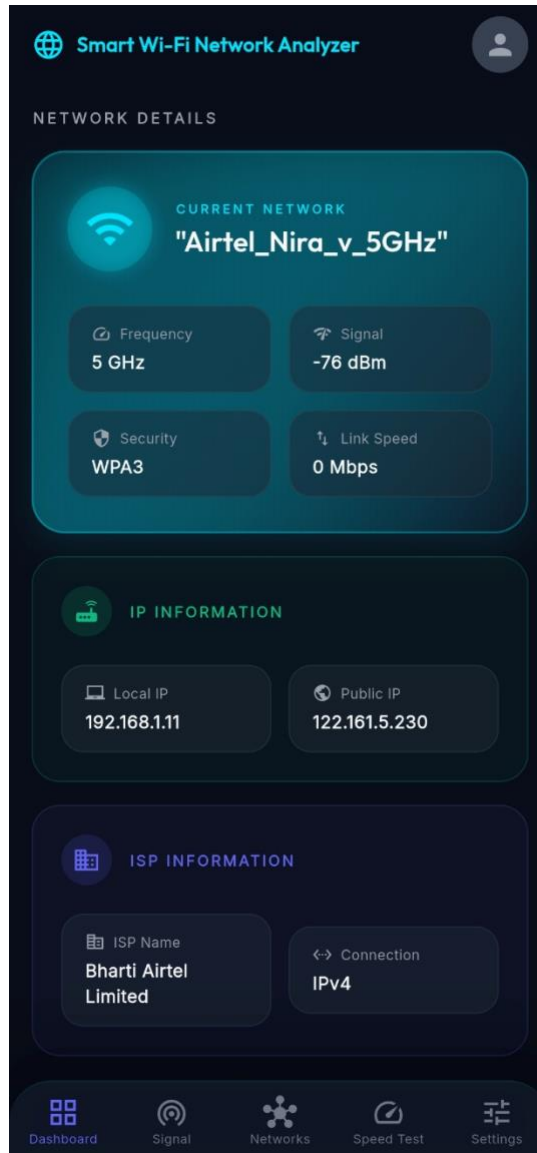
User interface testing ensured that the application layout is responsive, visually clear, and easy to use. The signal graph updates in real-time and provides a smooth visualization of network fluctuations. Stress testing was also performed by running the application continuously and executing multiple speed tests sequentially, during which no crashes were observed, though minor delays occurred after repeated operations due to network load.

Overall, the testing and validation results indicate that the Smart Wi-Fi Network Analyzer functions effectively as a network monitoring tool. While minor limitations exist in speed test accuracy, the application successfully provides reliable and real-time insights into Wi-Fi performance, making it suitable for practical use and further enhancement.

CHAPTER: 10

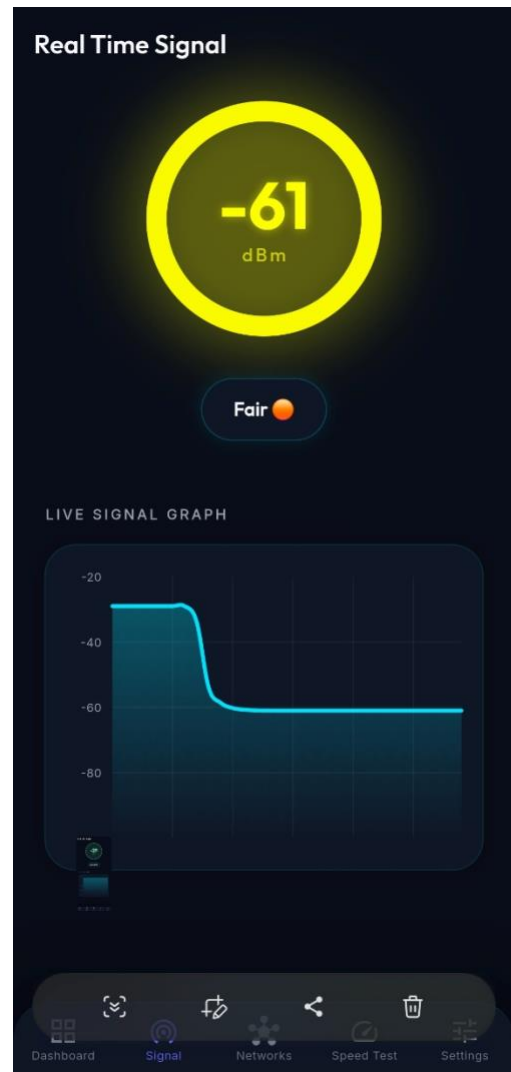
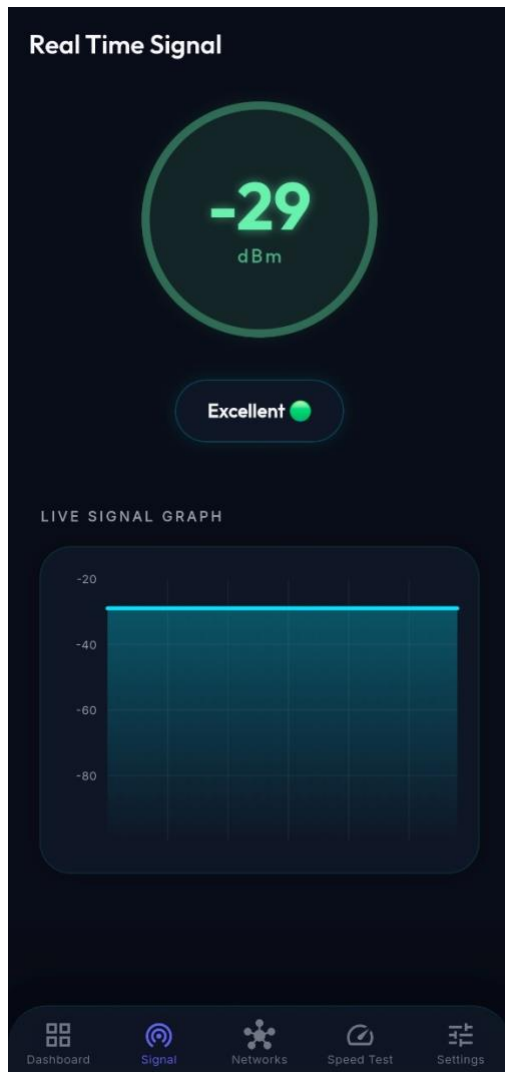
FINAL SOLUTION

Dashboard:

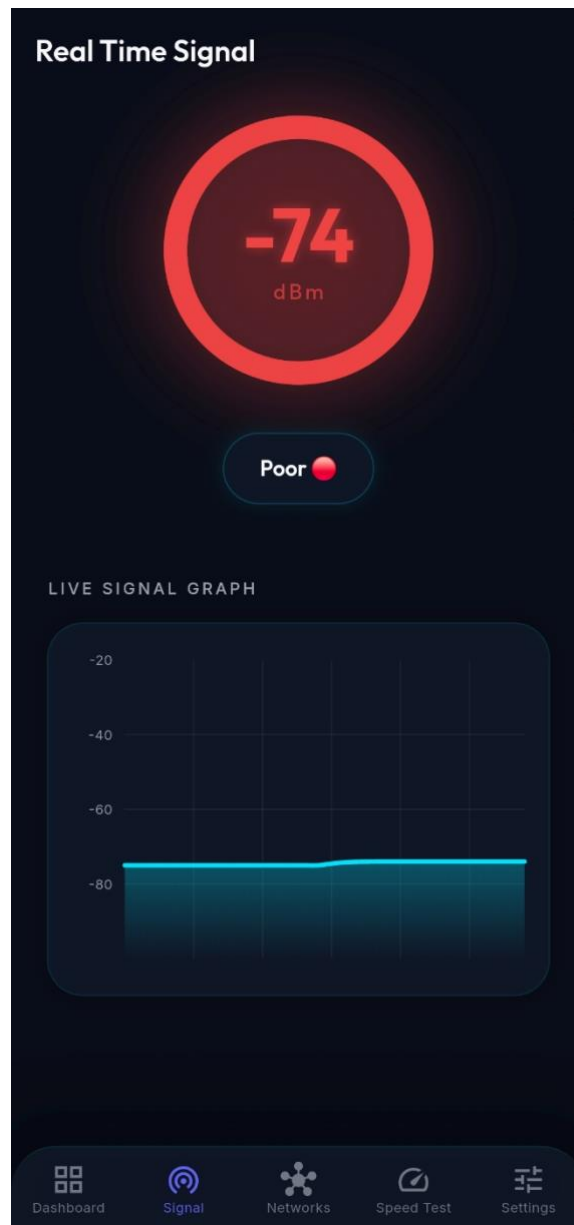


The dashboard of the Smart Wi-Fi Network Analyzer provides a clear and organized overview of the current network status. It displays the connected Wi-Fi name along with key parameters such as frequency, signal strength, security type, and link speed. Below this, the IP Information section shows both local and public IP addresses, helping users understand their network identity. The ISP Information section further provides details about the service provider and connection type. The interface uses a modern dark theme with card-based layout, making it visually appealing and easy to navigate, while the bottom navigation bar allows quick access to different features of the application.

Signal:

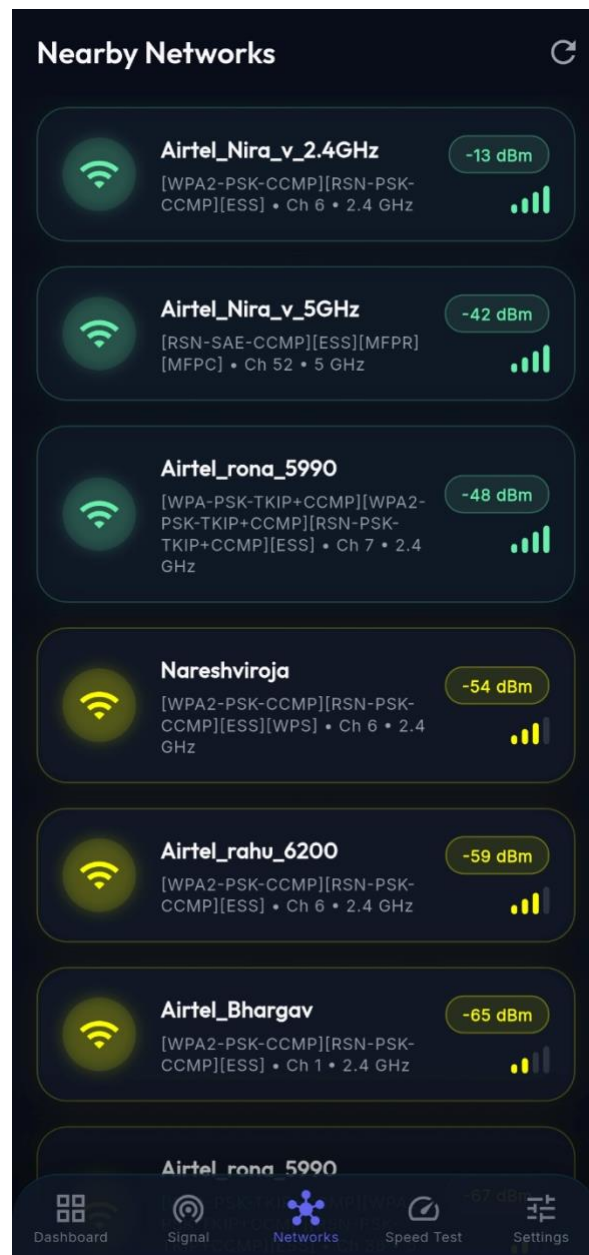


The Real-Time Signal feature of the Smart Wi-Fi Network Analyzer provides continuous monitoring of Wi-Fi signal strength using both numerical and graphical representations. The signal strength is displayed in dBm along with a circular visual indicator that changes colour based on signal quality. For example, in an excellent condition, the signal strength is around -29 dBm, shown in green, indicating strong and stable connectivity suitable for high-performance tasks such as streaming and gaming. In contrast, when the signal strength drops to around -61 dBm, the indicator changes to yellow and is labelled as “Fair,” representing moderate connectivity where users may experience reduced speed or occasional instability. Additionally, the live signal graph visually tracks fluctuations over time, helping users understand network behaviour and identify variations in signal strength effectively.



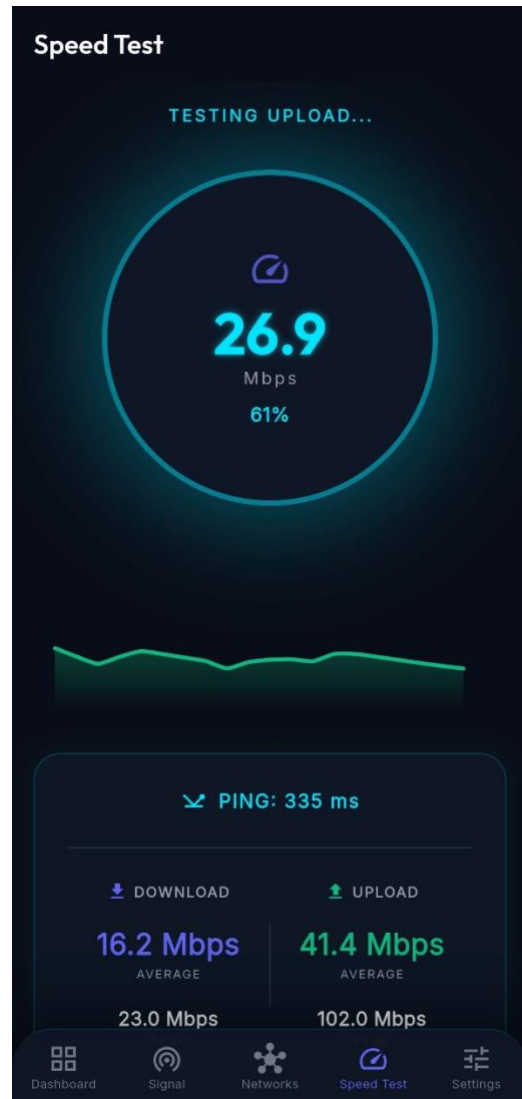
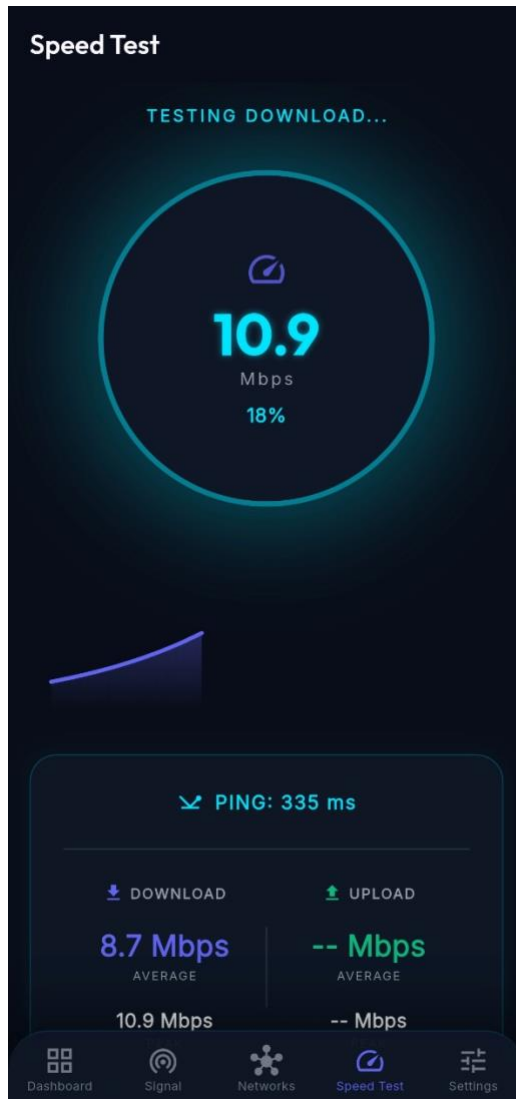
In the poor signal condition, the Wi-Fi signal strength drops to around -74 dBm, indicating weak and unstable connectivity. The circular indicator turns red, clearly representing low network quality, and is labeled as “Poor” to provide immediate user understanding. The live signal graph shows consistently low values with slight fluctuations, reflecting weak signal reception over time. In this condition, users may experience slow internet speeds, frequent disconnections, and difficulty in performing high-bandwidth activities such as streaming or video calls.

Nearby Networks:



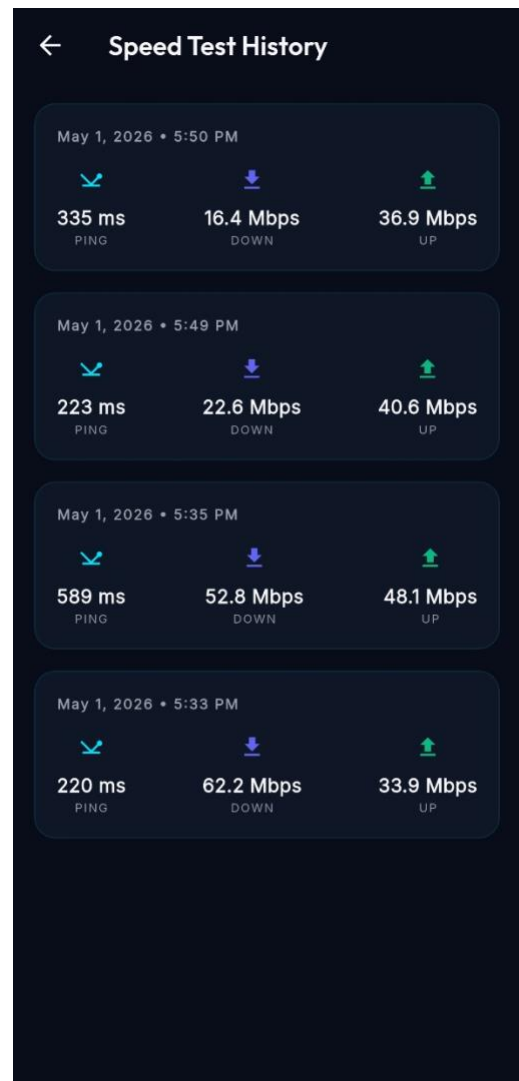
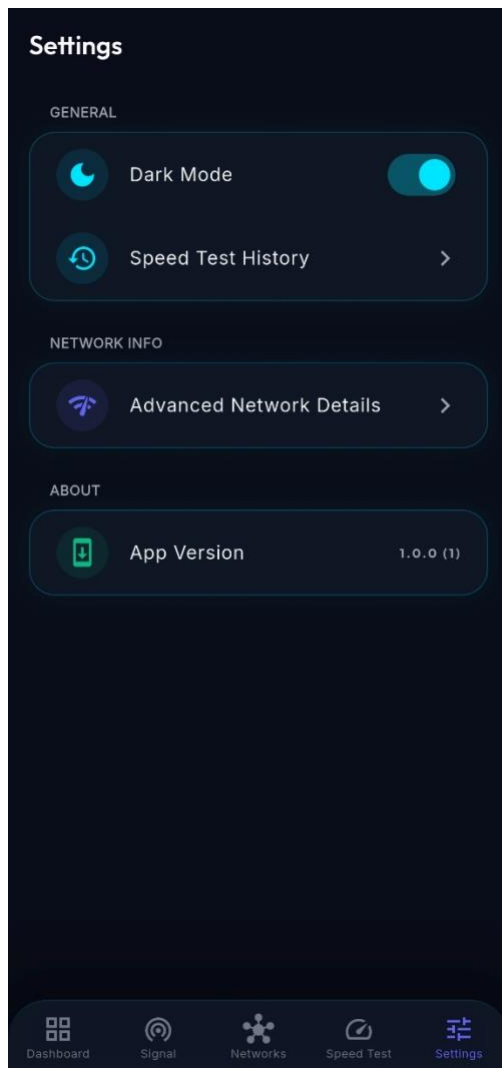
The Nearby Networks module displays a list of all available Wi-Fi networks within range, along with detailed technical information for each network. It shows the network name (SSID), security type, channel number, frequency band (2.4 GHz or 5 GHz), and signal strength in dBm. Networks are visually differentiated using colour indicators based on signal quality, helping users quickly identify strong and weak connections. This module enables users to analyse surrounding networks, detect interference, and choose the most optimal network for better connectivity and performance.

Speed Test:



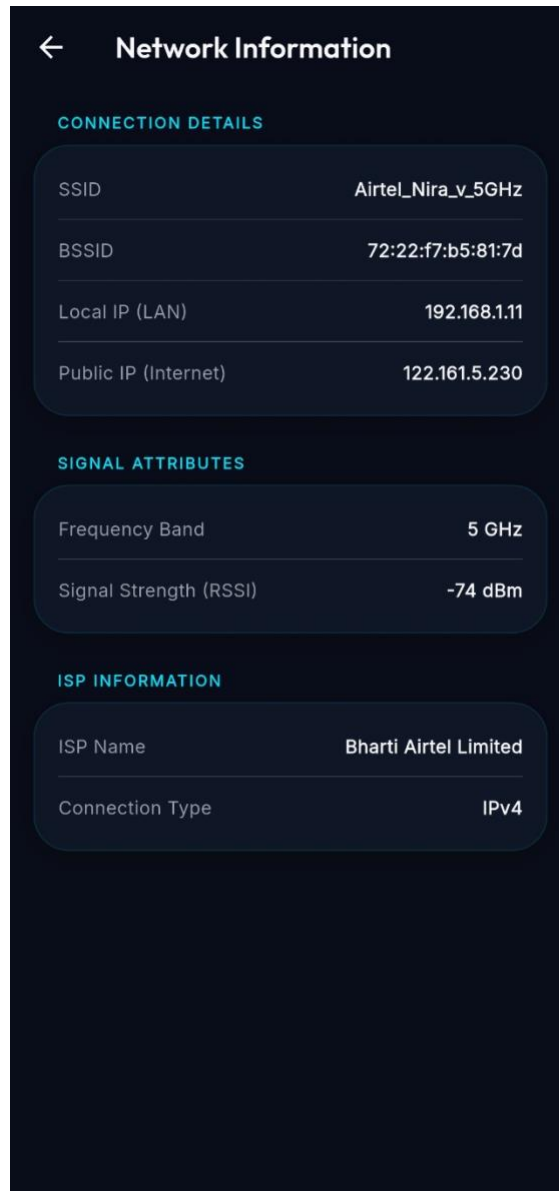
The Speed Test module measures the network performance by evaluating both download and upload speeds along with latency (ping). During the download phase, the application calculates how fast data can be received from the internet, while in the upload phase, it measures how quickly data can be sent to the server. The process is visually represented using a circular indicator showing real-time speed in Mbps along with progress percentage. Additionally, a graph displays speed variation during the test, and detailed statistics such as average and peak speeds are provided for both download and upload. This module helps users assess overall network performance and identify issues related to slow internet connectivity.

Settings:



The Settings module provides essential configuration options and additional utilities for the application. It allows users to enable or disable dark mode for better visual comfort and access the Speed Test History feature. It also includes an option for Advanced Network Details, which presents more detailed technical information about the current network connection. Additionally, the module displays the application version for reference and updates.

Within this module, the Speed Test History section stores previously recorded network performance results in a structured format. Each record includes the date and time of the test along with key metrics such as ping (latency), download speed, and upload speed. This enables users to track performance trends over time, compare results, and identify variations in network quality, making it easier to analyze and troubleshoot connectivity issues.



The Advanced Network Information module provides detailed technical insights about the current Wi-Fi connection. It includes connection details such as SSID, BSSID, local IP address (LAN), and public IP address, helping users identify their network and device within the system. The signal attributes section displays important parameters like frequency band and signal strength (RSSI), allowing users to evaluate connection quality. Additionally, the ISP information section provides details about the service provider and connection type. This module is useful for users who require deeper network analysis and troubleshooting capabilities.

CHAPTER: 11

CONCLUSION

The Smart Wi-Fi Network Analyzer project has been successfully designed and implemented as a comprehensive mobile application for monitoring and analysing Wi-Fi network performance. The system integrates multiple functionalities such as real-time signal monitoring, network information analysis, nearby network scanning, and internet speed testing into a single, user-friendly platform. This eliminates the need for multiple tools and provides users with a centralized solution for understanding and managing their network connectivity.

Throughout the development process, emphasis was placed on accuracy, usability, and real-time data visualization. The application effectively displays critical network parameters such as signal strength, IP details, ISP information, and performance metrics like download and upload speeds. The inclusion of features like live signal graphs and speed test history enhances the analytical capabilities of the system, allowing users to track performance trends and identify connectivity issues over time.

Although minor limitations exist, particularly in speed test accuracy due to implementation constraints, the overall system performs reliably and meets its intended objectives. The project demonstrates a strong understanding of networking concepts, mobile application development, and system-level data handling. It also provides a scalable foundation for future enhancements such as improved testing algorithms, cloud-based analytics, and advanced diagnostics.

In conclusion, the Smart Wi-Fi Network Analyzer serves as an effective and practical tool for real-time network monitoring and analysis, offering both technical insight and ease of use, and fulfilling the core requirements of the project successfully.